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FIRST SEMESTER M.Tech EXAMINATION, JANUARY -2024

INFERENTIAL STATISTICS AND DATA ANALYSIS

Time: 3 Hours **Maximum Marks: 100**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.
- **ii.** Missing data may be suitably assumed.

1. a. Calculate first four moments of mean from the following data. **10**

X	0	10	20	30	40	50	60	70	80
F	8	13	18	23	28	23	18	13	8

b. Explain the following with suitable examples: **10**

- **i.** Moments
- **ii.** Coefficient of skewness
- **iii.** Standard deviation
- **iv.** Quartile deviation
- **v.** Measure of dispersion

2. a. Find Bowley' s coefficient of skewness for the following values and interpret the results **10**

Size	4	4.5	5	5.5	6	6.5	7	7.5	8
f	10	18	22	25	40	15	10	8	7

b. Calculate mean deviation from mean and median for the following data: **10**

20, 70, 120, 170, 290, 420, 430, 530, 601.

Also calculate coefficients of M.D.

3. a. Describe various methods of sampling with suitable examples. **10**

b. **i.** Test the hypothesis that, the average content of containers of a particular lubricant is 10 litres. If the content of the random sample of 10 containers are 10.2, 9.7, 10.1, 10.3, 10.1, 9.8, 10.4, 10.3 and 9.8, use 1% level of significance. (Hint: $t_C = 3.250$). **10**

ii. An electrical firm manufactures light bulbs that have a length of life that is approximately normally distributed with a mean of 800 hours and a standard deviation of 40 hours. Test the hypothesis at 5% and 1% level of significance. $H_0: \mu = 800$ hours against $H_1: \mu \neq 800$.If a random sample of 30 bulbs has an average life of 788 hours. ($z_{0.01} = 2.85$ and $z_{0.05} = 1.96$)

OR

4. a. Following is the data collected from thirteen students before and after giving training on statistics and marks were recorded. Test the statistic whether there is a significant difference in the training. (t_{stat}: 2.1788) **10**

S. No	Post Training Score	Pre Training Score
1	82	78
2	74	70
3	45	49
4	48	48
5	56	62
6	89	88
7	74	78
8	45	40
9	48	45
10	56	50
11	51	58
12	80	75
13	40	50

b. The mean Mathematics score for the population of first-year students at university is 520. The standard deviation of scores in this population is 95. An investigator believes that the mean Mathematics score of first-year B.Tech course is significantly different from the mean score of the population. The mean of a sample of 36 first year B.Tech students is 548. Test the investigator prediction using 5% level of significance by one sample Z-test.(Critical Value:1.96) **10**

5. a. A supplier of CFL bulbs claims that out of all the CFL supplied, not more than 10% are defective. Sample of 300 bulbs tested, and 35 of them are defective. Can we conclude that the suppliers claim is true? **10**
(Critical Value: 1.96)

b. Based on field experiments, a new variety of black gram is expected to give a yield of 13 quintals per hectare. The variety was tested on 12 randomly selected farmer's fields. The yields (quintal/hectare) were recorded as 15.3, 14.6, 13.7, 11.9, 13.2, 13.1,11.4, 13.2, 12.6, 13.2, 14.1, and 13.1. Do the results confirm the expectation? (Critical value: 2.0201) **10**

OR

6. a. Fit a binomial distribution to the following data and test for its goodness of fit at 5% level of significance. **10**

x_i	0	1	2	3	4
f_i	419	352	154	56	19

(Critical value: 7.82)

b. To test the claim that the median age of mathematics faculty in the state community college atleast 42 years the results from a random sample of 32 mathematics faculty gave the following ages(in years) **10**

56 62 61 54 52 32 24 35 50 42 52 49 26 31 31 54

38 36 45 53 37 40 38 31 29 25 45 32 49 39 36 38

Use the sign test at 5% level of significance to test the claim by

i. critical values method

ii. p-values method

(Critical value: -1.645, $\phi(z = 0.71842)$ =0.2611)

7. a. The data below shows the salaries (in $) in randomly selected advertisements in two different occupation (Education & Marketing) **10**

Education(X)	22	40	18	25	15	23	16	19	21	30	
Marketing(Y)	28	24	20	45	50	39	26	55	48	41	42

Use Mann-Whitney test at 1% LOS to test that the median salary in the fields of education is lower than the median salary in the field of marketing.

(Critical value: -2.33)

b. Following data were collected which are achievement test score for four different rent teaching techniques. The objective of the experiment is to test the hypothesis of no difference in the population distributions of achievement test scores versus the alternative that they different in locations that is at least one of the distribution is shifted above the other. Conduct the test using Kruskal-wallis H-test with $\alpha = 0.05$. **10**

I	65	87	78	79	87	69	
II	75	69	83	81	72	79	90
III	59	78	67	62	83	76	
IV	94	89	80	88	96		

(Critical value: 7.814)

OR

8. a. Explain Maximum likelihood method and hence estimate the parameters of normal distribution using maximum likelihood method. **10**

b. i. A population of 29-year-old males has a mean salary of $29,321 with a standard deviation of $ 2,120. If a sample of 100 men is taken, what is the probability their mean salaries will be less than $29,000? ($\phi(1.51) = 0.4345$) **10**

ii. Suppose that the average resting heart rate for men over 19 years old is 70 beats per minute (bpm) with a standard deviation of 8 bpm. What is the probability that a randomly selected male 19 years old or older will have a resting heart rate of more than 83? ($\phi(1.63)$ =0.4484)